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Knowledge innovation network externalities in the Guangdong–Hong Kong–Macao Greater Bay Area: borrowing size or agglomeration shadow?

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ABSTRACT

The Guangdong–Hong Kong–Macao Greater Bay Area (GBA) is one of the most active innovation areas in China, but the complex institutional and cultural environment makes innovation cooperation among the cities challenging. Based on 2001–2019 data, this study analyses the spatial pattern and externalities of GBA's knowledge innovation network using a social network method and spatial econometric model. Results show that Guangzhou and Hong Kong have always been the cores of knowledge innovation in network. Shenzhen emerged as an innovation centre after 2012, and other cities have become peripheral areas in the network. Small- and medium-sized cities do not benefit from the innovative development of core cities but are trapped in their agglomeration shadow. Institutional and cultural differences are the main obstacles hindering innovation cooperation between cities. In comparison, distance has fewer limitations on innovation cooperation. The negative externality of knowledge innovation network indicates that this region should narrow the gap of spatial differences, optimise the innovation network pattern to improve the network externalities.

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1. Introduction

In the knowledge economy, urban agglomerations have become incubators for regional innovation. With the transition from 'space of place' to 'space of flow' (Castells 1996a, 1996b), the innovative spatial pattern of urban agglomerations has gradually evolved from traditional hierarchies to networks. Cities have transformed from independent points in space to hubs in networks, and interactions between cities have become important factors affecting innovation performance (Cooke et al. 2011).

The Guangdong–Hong Kong–Macao Greater Bay Area (GBA) is one of the regions with the highest degree of openness and the most economic vitality in China. To maintain its competitiveness, the Chinese government issued the 'GBA Planning Outline' in 2019, aiming to build the area into a world-class urban agglomeration and international innovation highland, which considers Guangzhou, Shenzhen, Hong Kong and Macao as four innovation centres.

However, it is unclear whether the innovation network of GBA has formed a multi-centre pattern, and what its network externality is. Most research on urban networks in GBA analyses from the perspective of economic factor flows (Qiu et al. 2019; Wang et al. 2020). Although some studies have examined innovation in this region, including exploring the innovation development path (Gu,

Cao, and Yang 2018), measuring the spatial structure of the innovation network and analysing its influencing factors (Ma and Zeng 2020), they have not further explored the network structure's external influence on regional innovation. We are therefore left to ask, what is the externality of the current innovation network pattern? Are cities surrounding innovation centres affected by borrowing size or agglomeration shadow?

Guided by these questions, this study measures the spatial pattern and externality of the GBA innovation network to clarify big cities' effects on regional innovation and provide useful references. Compared with other research on urban networks, this article not only analyses the network structure, but also studies its spatial effects. Compared with research on innovation, this paper analyses the innovation cooperation between cities and focuses on the spatial impact of central cities. The remainder of this paper is organised as follows. Section 2 provides a literature review on urban network externalities and innovation networks. Section 3 introduces the research scope, data and models. Section 4 describes the empirical analysis, and Section 5 discusses the optimisation and adjustment of urban network externalities in the GBA and concludes.

2. Literature review

2.1 From agglomeration externality to network externality

Agglomeration and diffusion are contradictory phenomena in the spatial evolution of urban agglomerations. The classic 'backwash-spread' theory, proposed by Myrdal, and 'trickle down' effect proposed by Hirschman (1958), both explain the imbalance of regional development. The spatial spillover effect of a central place gradually decreases under the constraint of distance. While driving the development of surrounding areas, the central place causes peripheral areas to fall into a trough (Zhang and Zhuang 2000) and form a 'core-periphery' structure. When Friedmann (1972) created the core-periphery theory, drawing on Schumpeter's innovative ideas, he believed that development could be regarded as a discontinuous accumulation process from basic innovation groups to large-scale innovation systems. Large cities with favourable conditions for innovation activities often become the core area of regional innovation, while the surrounding areas are reduced to a subordinate position; then, the two together form a complete innovation space system. Focusing on the relationship between cities, Alonso (1973) proposed the concept of 'borrowed size'. He pointed out that small cities close to metropolitan areas have higher per capita income levels than cities of the same scale and independent development. This is because these small cities can realise some of the advantages of large cities while avoiding their disadvantages by borrowing size. Krugman and Fujita (1995) introduced the concept of 'aggregation shadow', referring to the situation in which some small cities do not benefit from large neighbouring cities but are instead disadvantaged by them (Meijers, Burger, and Hoogerbrugge 2016).

Above theories have led many researchers to verify the two-sided externalities of agglomeration. Some scholars divided agglomeration economies into localised and urbanised economies and discussed their external effects on industries and urban economies (Tao et al. 2019; Wu, Wei, and Chen 2020). Other scholars have verified borrowing size as a positive agglomeration externality (Martinus et al. 2021). The essence of borrowing size is the complementarity between the local advantages of small cities (such as lower rents and less traffic congestion) and the advantages of big cities (such as a broad market, labour pool and diverse consumption/cultural facilities) (Phelps, 1998). However, some scholars have also found that big cities have an agglomeration shadow effect on small cities. Specifically, there are significant differences in land use types between metropolises and their shadow areas (Wu, Zhou, and Liu 2020). Per capita income in large cities is much higher than surrounding areas, resulting in imbalances in the distribution of labour and the spatial pattern of the economy (Au and Henderson 2006; Sun et al. 2019). A 2004 research report by the Asian Development Bank introduced the 'Beijing-Tianjin Poverty Belt', which refers to a C-shaped belt poverty area around Beijing and Tianjin (Luo and Ma 2015). This prompted scholars

to research the formation mechanism of the agglomeration shadow and poverty reduction measures (He, Fan, and Li 2018).

The progress of technology and the advent of the network society have made various factors, such as population, capital, information etc., flow between cities convenient and frequent, and mechanisms of interaction between cities are also changing. Cities and the companies within them not only benefit from local exchanges, but also increasingly affected by external regions (Wang et al. 2020). Access to agglomeration advantages is sufficient to the company's operation without having to exist in or be close to the agglomeration area (Speldekamp, Knobben, and Saka-Helmhout 2020). However, the perspective of agglomeration externality, based on the centre periphery theory, cannot explain the agglomeration phenomenon on a larger scale because it confines the city's influence on its neighbouring and ignores cross-regional connections (Fan et al. 2021). In view of this, Castells (1996a, 1996b) introduced the concept of 'space of flows', referring to the spatial pattern formed by the flow of various elements. Taylor, Hoyler, and Verbruggen (2010) proposed the central flow theory based on the external urban relations to criticise central place theory based on interurban hierarchical relations. In the context of space of flow, the intercity flow of various elements has shaped connections between cities, shifting the study of regional spatial structure from the internal attributes of cities to the external connections (Inostroza and Zepp 2021).

Capello introduced 'network externality' and found that it improves with an increase of nodes in the network. By participating in network, cities can achieve functional complementarity through cooperation, obtain economies of scale and reduce uncertainty. By joining the network, cities can obtain two effects (Meijers 2005): (1) the horizontal synergy effect, which refers to the scale effect brought about by simple cooperation between cities; (2) vertical synergy effect, which refers to the division of labour and specialised production between cities. In this context, the realisation of borrowing size is no longer based on geographical proximity but through embedded in networks (Meijers, Burger, and Hoogerbrugge 2016). Therefore, a city's ability to establish a network connection has become important in realising the borrowing size.

In general, the agglomeration economy causes externalities. No matter how the regional spatial structure changes, borrowing size and agglomeration shadow together comprise the pros and cons of externalities (Meijers, Burger, and Hoogerbrugge 2016). The final effect of externalities depends on the tradeoff between the two effects. Therefore, analysing the role of externalities is of great significance for objectively evaluating and optimising regional spatial structures.

2.2 Innovation network externality

Generally, the development of innovative activities begins in a non-equilibrium state due to the heterogeneous distribution of innovative resources (Mohnen, DeBresson, and Shiqing 2002). Under the effect of path dependence, innovation resources continue to be highly concentrated in a certain region ultimately making it an innovation pole. While areas with less innovative resources are innovation holes, which together with the innovation pole constitute the spatial pattern of innovation (Gonçalves, de Matos, and de Araújo 2019). As the connections and interactions involved in innovation continue to strengthen within the innovation pole, innovation networks are generated and continuously expanded. Driven by the promotion of globalisation and knowledge economy, innovation activities continue to expand, eventually forming an innovation network that brings together local and cross-regional interactions (Fan, Lian, and Wang 2020). This type of network structure has the characteristics of coexistence, with obvious agglomeration boundaries and openness, making the spatial effect of the innovation network uncertain.

From the perspective of internal factors, the structure of the innovation network significantly affects its effectiveness (Taalbi, 2020). The density and centrality of network are positively correlated with the innovation performance (Bai et al. 2021). Enterprises embedded in innovation network have more innovation output than those outside the network (Fan, Lian, and Wang 2020). The innovation cooperation relationship can be divided into strong and weak connections.

Strong connections can enhance trust and cooperation and promote resource flow and knowledge sharing within the network (Hansen 1999). However, some scholars believe that strong connections will reduce the diversity of members in innovation network, thereby reducing the sources of new ideas and technologies. A weak connection has low-cost advantages and higher openness, which can extend the connection outside the network and promote cross-border communication (Baron and Gomez 2013). Rowley, Behrens, and Krackhardt (2000) found that strong ties encourage incremental technological innovation, while weak ties encourage breakthrough technological innovation. Capaldo (2007) emphasised innovation networks with only strong or weak connections are insufficient for producing external effects, and a two-layer network structure with strong internal structures and weak external connections should be constructed. Eisingerich, Bell, and Tracey (2010) measured innovation network structure from two aspects of network strength and openness and found that network strength's positive effect on innovation performance decreases with increasing environmental uncertainty, while the positive effect of network openness is positively correlated with environmental uncertainty. These views mean that the external environment should be considered when discussing the innovation network effectiveness.

From the perspective of external factors, Boschma and Frenken (2010) demonstrated that cognitive proximity, social proximity, organisational proximity, institutional proximity and geographical proximity are the main factors affecting the spatial evolution of innovation network. Except that the spatial effect of network attenuates with distance increasing (Foxon 2014), the spatial concentration of innovation is likely to form a stable and rigid closed relationship network, resulting in a negative effect of geographical proximity 'lock-in', which is unfavourable to innovation (Bathelt, Feldman, and Kogler 2012). However, some scholars believe that the development and popularisation of the Internet will lead to 'a flat world', thus redefining the spatial network in terms of soft innovation (Howells and Bessant 2012). Social and cultural factors are playing increasing roles on innovation. There will be more interaction and trust between innovative members with closer cultural customs, thereby promoting innovative cooperation (Porter 1998). However, the excessive singularity of cultural customs may produce xenophobia and discourage innovation diversification. Furthermore, institutional proximity reflects the degree of similarity of enterprises constrained by formal or informal rules. On the one hand, it can reduce the uncertainty and conflict of innovation cooperation (Balland, Belso-Martínez, and Morrison 2016). On the other hand, excessive institutional proximity may lead to excessive dependence on organisations, threaten the protection of intellectual property rights and reduce innovation incentives (Molina-Morales et al. 2015).

With the advancement of globalisation and technology, innovation cooperation can break through geographical boundaries and expand outward. High density and openness have become the source of spatial effect of the innovation network, and soft environment factors, such as culture and institution, impose significant influences. In China, GBA is one of the most vigorous innovation regions. However, the special environment of 'one country, two systems and three customs areas' means its innovation faces more cultural and institutional challenges than other regions. The Chinese government plans to establish a polycentric innovation network centred on Guangzhou, Shenzhen, Hong Kong and Macao in here. What is the current situation of innovation in this region? Is the externality of innovation network a positive borrowing size or negative agglomeration shadow? Previous studies have focused on the evolution of network spatial structure (Li, Li, and Qiu 2019; Ma and Zeng 2020), innovation performance (Wang, Zhang, and Fan 2019; Fan and Lian 2021) and influencing factors (Wen, Deng, and Chen 2017), while ignoring network externalities, so they cannot find out the truly spatial effect of innovation networks. In view of this, this study focuses on analysing the externalities of innovation network under the influence of distance, cultural and institutional differences to clarify the impact of complex external environments on network effects and make up for the existing literatures.

3. Data and methods

3.1 Research scope

The GBA was first formally proposed in China's plan for the 'One Belt and One Road' in 2015 and was subsequently included in the national '13th Five-Year Plan' in 2016. The GBA consists of two special administrative regions of Hong Kong and Macau and nine cities of Guangzhou, Foshan, Zhaoqing, Jiangmen, Zhongshan, Zhuhai, Shenzhen, Dongguan and Huizhou in Guangdong Province. It operates two political systems and three separate customs territories, giving it a unique and complex soft environment different from other regions in China and bay areas around the world (Figure 1).

3.2 Data sources

Indicators of innovation include papers and patents. Paper data can be used to reflect knowledge innovation, while patent data is more used to represent technological innovation. Therefore, this study uses the number of all research papers published in city to reflect its knowledge innovation. Correspondingly, the collaborative paper between cities can be used to measure the intercity knowledge innovation cooperation, and papers without cooperation with other cities reflect the city's independent knowledge innovation ability. Above three types of paper data comes from the 'web of science'. The study period is set to 2001–2019, the reasons are as follows (1) Hong Kong and Macau became the two special administrative regions of China in 1997 and 1999 respectively, and China joined the World Trade Organization in 2001, all of which greatly promoted cross-border cooperation. (2) Although it was not until 2019 that the Chinese government issued the plan of GBA, the local governments had already conceived and promoted cooperation at the beginning of the twenty-first century. Therefore, it is reasonable to set the starting year as 2001.

We set up two dummy variables to measure the impact of the GBA's cultural and institutional environment on innovation cooperation. First is the cultural difference dummy variable ($culture_{ij}$),



Figure 1. Map of Guangdong–Hong Kong–Macao Greater Bay Area.

which uses cities' dialects to represent cultural differences. Dialect is divided into three levels from large to small: dialect region, dialect district and dialect zone. Cultural difference dummy variables are set from 0 to 3 according to the different levels of dialects in every two cities.

Second is the institutional difference dummy variable (institution_{ij}), which reflects the differences in political systems, customs territories and administrative hierarchy of cities. There are two political systems in the GBA: the socialist system is implemented in nine cities in Guangdong province, and the capitalist system is implemented in Hong Kong and Macao. There are three customs territories: nine cities in Guangdong Province have the same customs territory, Hong Kong and Macau have their own customs territories. The administrative levels of cities are divided into three levels: Guangzhou and Shenzhen are sub-provincial cities, other seven cities are prefecture-level cities, Hong Kong and Macau are special administrative regions (equivalent to provincial level). Accordingly, the dummy variables of institutional differences between every two cities are set from 1 to 4.

This study also uses socio-economic indicators as control variables to measure the innovation network externalities, which all come from statistical yearbooks of Guangdong Province and each city.¹

3.3 Methods

3.3.1 Social network analysis

This study uses the social network analysis method and UCINET software to calculate network centrality and density. The indicator of degree centrality reflects the status and influence of a city in the network, which is the sum of the direct connections between one node and other nodes. Density reflects the closeness of innovation cooperation between cities. The knowledge innovation network in this study is an undirected network. Therefore, the relative degree centrality and network density formula are as follows:

$$\text{Degree centrality} = d(i)/(n - 1) \quad (1)$$

where $d(i)$ is the number of relationships owned by any node i in the network, and $(n - 1)$ is the maximum number of relationships that a node has in the network.

$$\text{Density} = m/n(n - 1)/2 \quad (2)$$

The m represents the total number of relationships contained in the network, and the denominator, $n(n - 1)/2$, is the maximum number relationships of theoretically possible in the network.

3.3.2 Spatial econometric model

We use the spatial econometric model to measure network externalities. We generate three spatial weight matrixes based on distance between every two cities, cultural and institutional differences dummy variables. Formulas are as follows:

$$W_{\text{dis}} = \begin{cases} \text{distance}_{ij}, & i \neq j \\ 0, & i = j \end{cases} \quad (3)$$

$$W_{\text{ins}} = \begin{cases} \text{institution}_{ij}, & i \neq j \\ 0, & i = j \end{cases} \quad (4)$$

$$W_{\text{cul}} = \begin{cases} \text{culture}_{ij}, & i \neq j \\ 0, & i = j \end{cases} \quad (5)$$

where distance_{ij} is the geographic distance between cities i and j , institution_{ij} and culture_{ij} are the institutional and cultural differences dummy variables of the two cities.

Since both the LM and robust LM results pass the 1% significance test, the following spatial Dubin model is used for empirical testing:

$$\text{paper}_{i,t} = \alpha_{i,t} + \beta_1 \text{centrality}_{i,t} + \delta X_{i,t} + \theta W \text{centrality}_{i,t} + \varepsilon_{i,t} \quad (6)$$

where $\text{paper}_{i,t}$ represents the number of published papers of city i in year t , including three categories: total papers, collaborative papers and local papers. W is the spatial weight matrix, including the distance, cultural and institutional spatial weight matrix. $\theta W \text{centrality}_{i,t}$ is the spatial lag of the centrality of city i in year t , and its coefficient θ can reflect the influence of the city's position in the network on the innovation development of surrounding areas. α is a constant term, $\beta\delta$ and θ are all parameters to be estimated, and ε is an error term.

To further explore the externalities of knowledge innovation networks, we examine the relationship between cities' centralities and regional differences to determine whether the development of urban innovation network can help narrow the gap between regions. The specific model is as follows:

$$\text{gap_paper}_{ij} = \alpha + \gamma_1 \text{gap_centrality}_{ij} + \varphi \text{gap_}X_{ij} + \varepsilon \quad (7)$$

where gap_paper_{ij} represents the knowledge innovation gap between cities i and j , $\text{gap_centrality}_{ij}$ is the gap of centralities between city i and j , and $\text{gap_}X_{ij}$ is the difference in control variables between the two cities.

In a nutshell, this article combines social network methods and spatial measurement models to test the externalities of knowledge innovation networks. By taking centrality as the core indicator to reflect city's status in network and testing its relationship with intercity differences, we not only examine the network's attributes, but also analyse its spatial effects, which helps to reveal the nature of knowledge innovation network and discover its problems, so as to provide useful suggestions for improving network externalities.

4. Results and analysis

4.1 Knowledge innovation network spatial pattern

We examined the knowledge innovation development trend in the GBA using the total papers, local papers and intercity collaborative papers. As Figures 2–4 show, the total number of papers published in the GBA from 2001 to 2019 kept increasing. Before 2011, Hong Kong ranked first in total papers. In 2013, Guangzhou's total and local papers both exceeded Hong Kong; Guangzhou maintained first in the cooperative papers throughout the study period. After 2012, Shenzhen ranked third in total papers, and the gap between it and Hong Kong continued shrinking. However, the number of papers in Macao has been relatively small. Therefore, Guangzhou, Hong Kong and Shenzhen are the three central cities in the current knowledge innovation network, while Macao has not yet shown a high central position in knowledge innovation.

The results of social network analysis show that the overall density of GBA's knowledge innovation network improved from 2001 to 2019, indicating that the innovation cooperation connections among cities are increasing (Figure 5). The overall centrality was calculated using the total number of papers published in each city. Hong Kong maintained the highest before 2012, was surpassed by Guangzhou after 2013, and then shows a declining trend. Guangzhou showed a rapid growth before 2012, but its growth trend slowed after 2013. Shenzhen surpassed Zhongshan in 2013 and ranked third. In 2019, Shenzhen's overall centrality was close to that of Hong Kong. The overall centrality of other cities is relatively low (Figure 6).

Cooperative centrality was calculated using the cooperative papers between every two cities. Guangzhou maintained the highest cooperative centrality. Zhongshan ranked second before 2018, but behind Hong Kong in 2019. Shenzhen ranked fourth in the GBA (Figure 7).

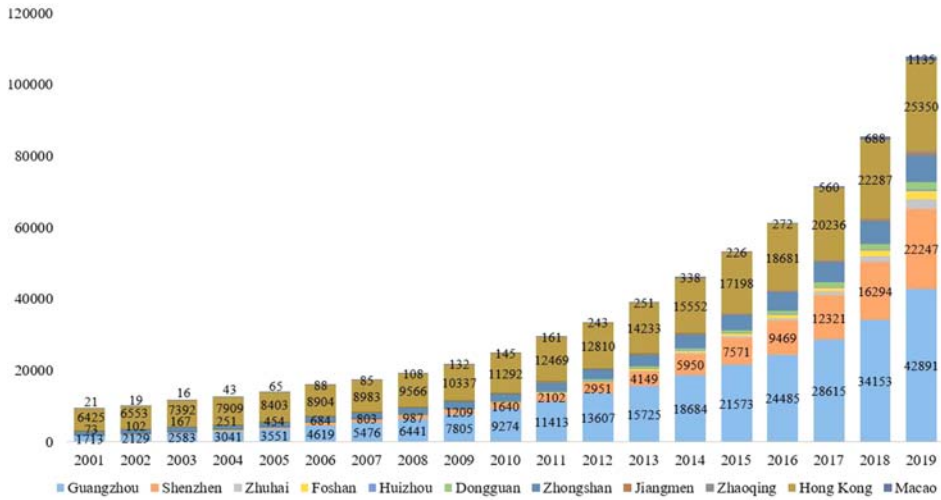


Figure 2. Total papers published in the GBA from 2001 to 2019.

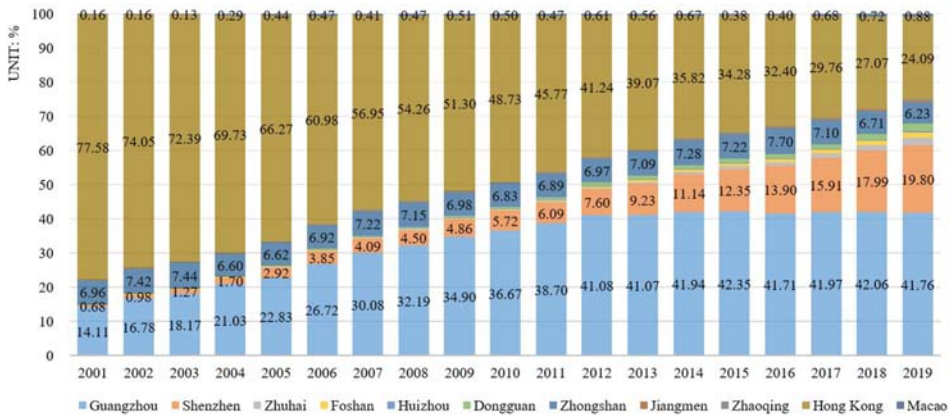


Figure 3. Proportion of local papers in each city from 2001 to 2019.

From the situation of overall and cooperative centrality, we see Guangzhou has been the core city in the network since 2013. Hong Kong and Shenzhen both have strong innovation capacities but their cooperation with other cities has been relatively limited. Macau is relatively weak in overall innovation and intercity cooperation.

Finally, Figure 8 shows the spatial evolution of the knowledge innovation network. It changed from a triangular structure centred on Hong Kong, Guangzhou and Zhongshan in 2001, to a four-link structure featuring Guangzhou, Hong Kong, Shenzhen and Zhongshan in 2010. In 2019, a diamond structure with Guangzhou, Shenzhen and Zhongshan as three centres and Zhongshan, Dongguan and Foshan as the sub-centres was formed.

4.2 GBA innovation network externalities

4.2.1 Innovation cooperation externality

The spatial effects of the knowledge innovation network pattern on overall, cooperation and independent innovation were measured respectively (Table 1). First, columns 2–4 of Table 1 shows the

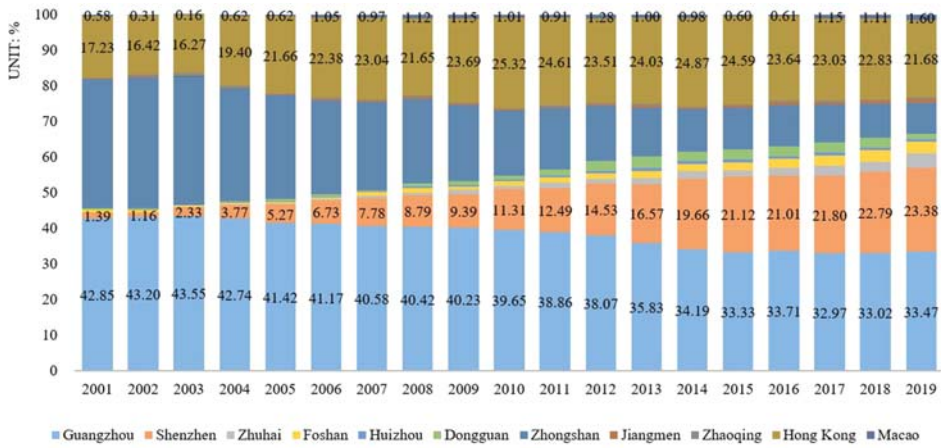


Figure 4. Proportion of collaborative papers in each city from 2001 to 2019.

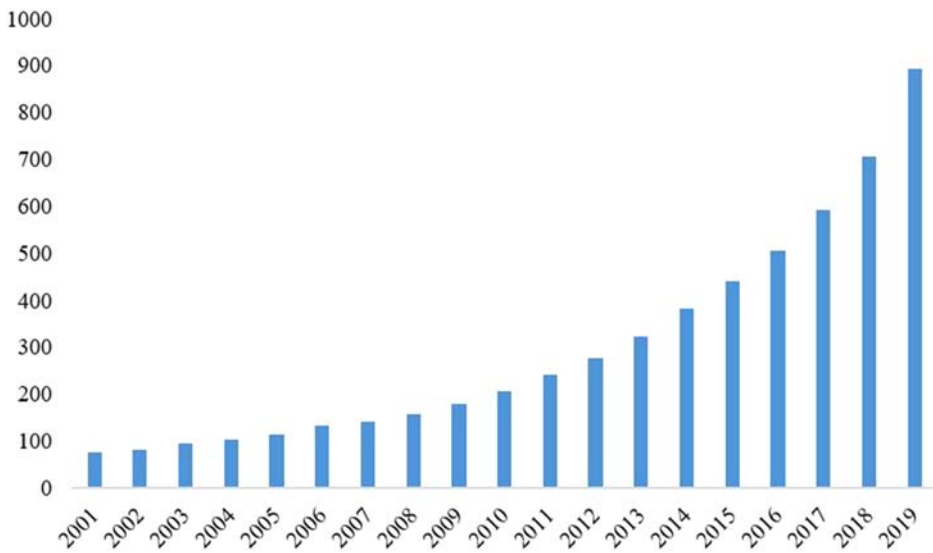


Figure 5. Knowledge innovation network density from 2001 to 2019.

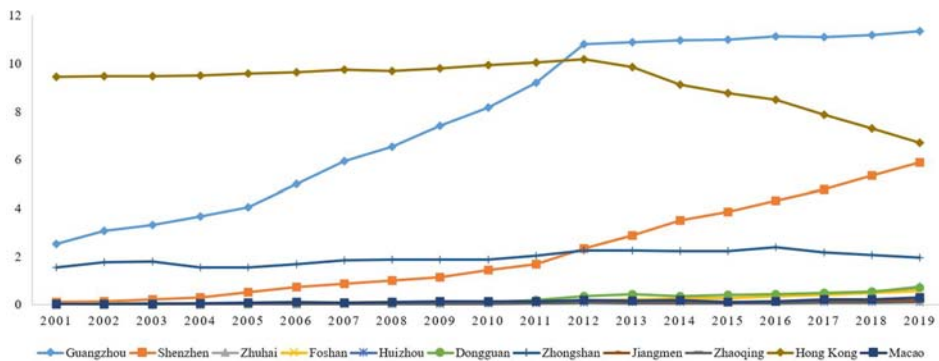


Figure 6. Overall centrality of knowledge innovation in each city from 2001 to 2019.

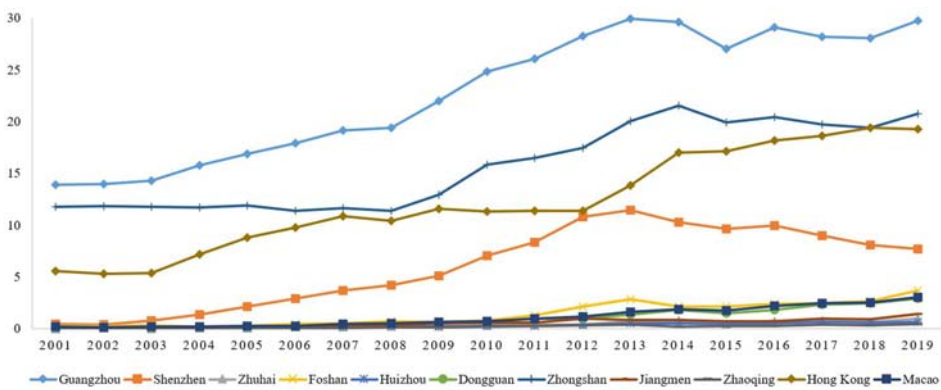


Figure 7. Cooperative centrality of knowledge innovation in each city from 2001 to 2019.

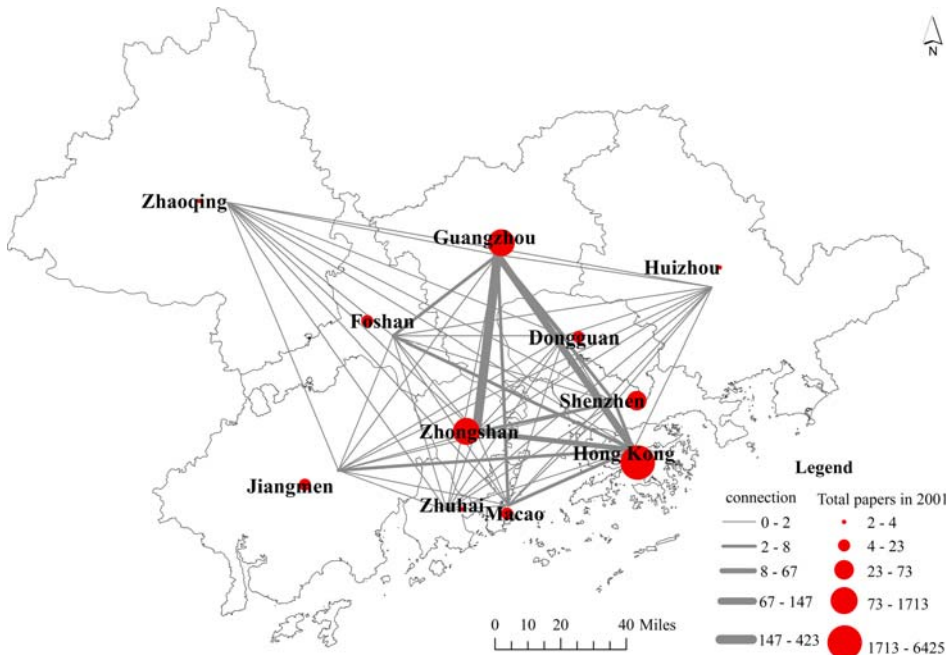


Figure 8 . (a) GBA knowledge innovation cooperation network in 2001. (b) GBA knowledge innovation cooperation network in 2010. (c) GBA knowledge innovation cooperation network in 2019.

results of network pattern on urban overall innovation. Under the influence of distance, the city's cooperative centrality has no impact on its own overall innovation, but has a significant negative impact on the surrounding areas. Under the impact of institutional differences, the urban cooperative centrality has an adverse effect both on itself and other cities. The impact of cultural differences is relatively weak, and the degree of urban cooperative centrality has no significant impact either on itself or the surrounding areas. This shows that the knowledge innovation network in GBA does not have positive externalities. More specifically, cities with high centrality negatively impact innovation in surrounding areas, and the adverse effects on cities with large institutional differences are more significant.

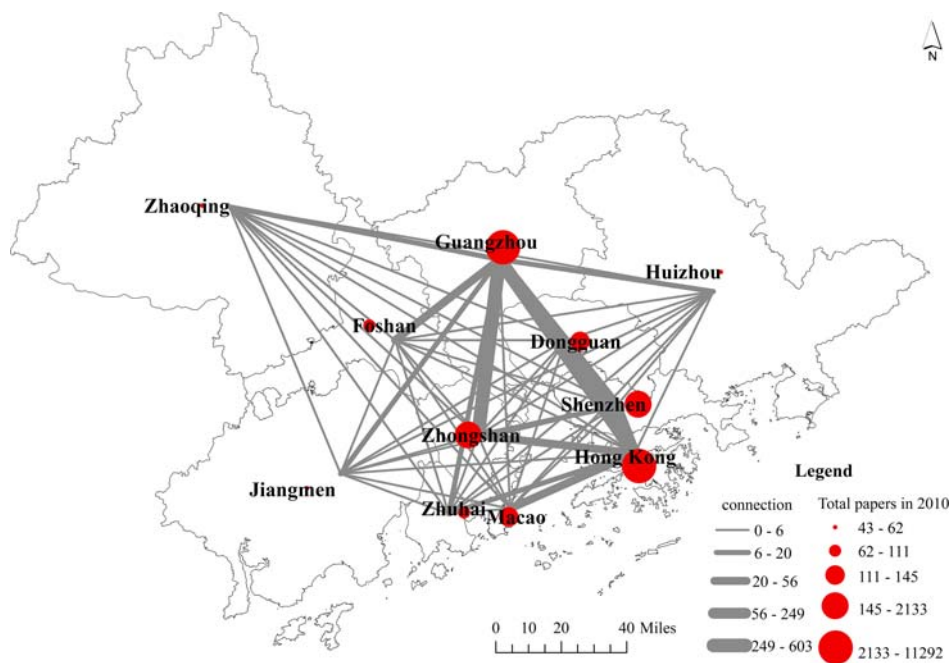


Figure 8 Continued

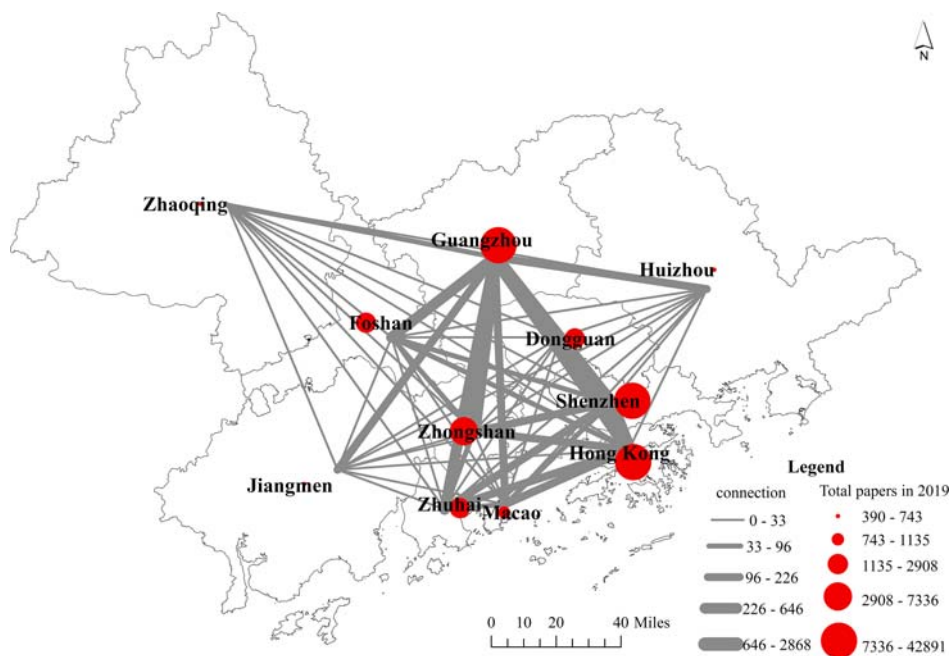


Figure 8 Continued

The results of the control variables show that the quality of urban researchers positively affects local knowledge innovation, and it becomes insignificant when cultural differences are considered. Whether under the influence of distance, institutional differences or cultural differences, research

Table 1. Results of the externalities of knowledge innovation network.

Explained variable	Overall innovation ability			Cooperative innovation ability			Independent innovation ability		
	Distance	institution	culture	Distance	institution	culture	Distance	institution	culture
Centrality	-0.12 (-1.48)	-0.14* (-1.84)	-0.03 (-0.32)	-0.15 (-1.54)	-0.17* (-1.73)	-0.57** (-2.27)	-0.10 (-1.32)	-0.11* (-1.69)	-0.14** (-2.08)
Lnresearcher	0.09** (2.02)	0.09* (1.90)	0.07 (1.38)	0.09* (1.80)	0.08 (1.53)	-0.90** (-2.38)	0.09** (2.08)	0.08* (1.84)	0.10*** (2.89)
Lnexpend	0.05* (1.82)	0.04* (1.92)	0.03 (1.27)	0.05 (1.59)	0.05* (1.72)	0.22** (2.58)	0.05** (2.13)	0.05 (1.14)	0.04 (0.76)
Ingdp	1.49*** (6.49)	1.28*** (3.64)	1.27*** (3.04)	1.07*** (3.84)	0.77* (1.90)	0.28 (1.14)	1.48*** (7.58)	1.27*** (4.13)	-0.14 (-0.34)
Popuflow	-0.03** (-2.30)	-0.03 (-1.42)	-0.03* (-1.70)	-0.02 (-1.35)	-0.02 (-1.00)	-0.01 (-0.44)	-0.02*** (-2.49)	-0.02 (-0.91)	-0.01*** (-2.96)
Internet	0.11** (2.31)	0.07** (2.30)	0.05 (1.38)	0.08 (1.34)	0.05 (1.12)	0.00 (0.64)	0.11*** (2.83)	0.07** (2.46)	-0.00 (-0.68)
W*centrality	-0.91** (-2.13)	-0.89* (-1.96)	0.44 (0.72)	-0.83* (-1.74)	-0.91* (-1.74)	-4.19** (-2.04)	-0.93*** (-2.66)	-0.82** (-2.06)	-1.72*** (-2.73)
Spatial rho	0.27** (2.32)	-0.25** (-2.28)	0.09 (1.11)	0.49*** (4.23)	0.12 (1.23)	-2.97*** (-4.73)	0.27** (2.67)	-0.38** (-2.50)	-1.50*** (-3.56)
R-sq	0.66	0.69	0.57	0.57	0.53	0.68	0.68	0.71	0.57

Notes: Standard errors are in parentheses. ***, ** and * indicate significant at the levels of 1%, 5%, and 10%, respectively.

funding positively impacts local innovation, and regions with higher economic levels tend to have stronger innovation capabilities. Furthermore, the relationship between population mobility and urban knowledge innovation is negatively correlated. The Internet penetration rate has a significant positive impact on knowledge innovation.

Second, Table 1, columns 5–7 show the network structure's influence on innovation cooperation between cities. The farther the distance between cities, the less cooperation occurs between cities. Institutional factors negatively affect cooperation both in local and other cities. This shows that urban knowledge innovation cooperation is mainly established between cities under the same institutional environment, and the greater the institutional difference, the less cooperation occurs. The impact of cultural differences is similar to the institutional differences. Therefore, institutional and cultural differences are the main obstacles hindering knowledge innovation cooperation between cities. Although distance limits cooperation, its influence is relatively small compared to institutional and cultural differences.

Third, we test the influence of urban innovation network structure on city's independent innovation (Table 1, columns 8–10). Under distance, institutional and cultural differences, urban centrality not only negatively affects local innovation, but also discourages independent innovation development of surrounding areas, and finally restrict overall urban innovation development. This further proves that the GBA's knowledge innovation network has negative externalities.

4.2.2 Intercity innovation gap externalities

Taking the difference in the total number of papers between every two cities as the explained variable to further explore the effects of innovation network pattern on the regional innovation gap (Table 2). There is a significant positive correlation between the gap in urban centrality and innovation capabilities, indicating that the higher the concentration of innovation network, the greater the innovation gap between cities. Distance and institutional differences aggravate the negative externalities of innovation centres, while cultural differences have relatively little impact. In addition, the gap in the Internet development level between cities also widens the innovation gap between cities.

4.3 Robustness and endogenous tests

The possible endogenous problem of above models is that there may be a causal relationship between the city's cooperative centrality and its knowledge innovation capabilities. Specifically, the more knowledge innovation cooperation, the more knowledge innovation output; on the contrary, the stronger the knowledge innovation ability of cities, the more innovative cooperation relationships there will be. In view of this, we use the lag term of cooperative centrality as an instrumental variable to revise the model. Results are shown in Table 3. The lagging term of urban cooperative centrality also has a negative impact on urban knowledge innovation. Under the distance spatial weight matrix, the negative impact of centrality is relatively small and insignificant. Under the spatial weight matrix of cultural and institutional differences, cooperative centrality both have a significant negative impact on cities' overall, and cooperative innovation. These results are consistent with the previous results, indicating that they are robust and reliable.

The above analysis reveals that, in the knowledge innovation network of GBA, central cities have significant negative externalities to other cities. The reasons for this phenomenon may be the following. First, the large difference in initial level of innovation between cities essentially determined the imbalance of network. In the development of Shenzhen, policy support has played a great role, while small and medium-sized cities do not enjoy the same favourable terms, which intensifies the resource plundering and development gap of the central city to the surrounding areas. Second, cities with higher degree of centrality are more tolerant of cultural and institutional differences than small and medium-sized cities and are leading in cultural diversity and institutional reforms. Therefore, a more superior soft environment for knowledge innovation gives the central city a

Table 2. Results of externalities on the intercity innovation gap.

Explained variable Model	The gap in knowledge innovation between cities				
	Model 1	Model 2	Model 3	Model 4	Model 5
gap_centrality	1.0007*** (3442.34)	1.0001*** (1932.15)	0.9979*** (859.12)	1.0002*** (1282.59)	0.9958*** (550.79)
dis*centrality		0.0006* (2.10)			0.0010 (1.48)
ins*centrality			0.0013** (2.74)		0.0019** (2.56)
cul*centrality				0.0003 (0.83)	−0.0001 (−0.17)
gap_gdp	0.0000 (−1.46)	0.0000** (−2.43)	0.0000 (−1.33)	0.0000 (−1.51)	0.0000** (−2.41)
gap_inresearcher	−0.0026 (−0.92)	−0.0021 (−0.79)	−0.0009 (−0.40)	−0.0026 (−0.92)	0.0008 (0.90)
gap_expend	0.0076 (0.65)	0.0100 (1.58)	0.0062 (0.60)	0.0073 (0.81)	0.0098** (2.67)
gap_popouflow	0.0018 (0.45)	0.0010 (0.37)	0.0006 (0.15)	0.0022 (0.69)	−0.0014 (−1.30)
gap_internet	0.0804** (0.0804)	0.1071** (2.51)	0.1092*** (3.55)	0.1101 (1.70)	0.1574*** (3.17)
cons	0.0035 (1.10)	0.0051* (1.95)	−0.0007 (−0.30)	0.0045 (1.72)	−0.0003 (−0.29)

Standard errors are in parentheses. ***, **, and * indicates significant at the levels of 1%, 5%, and 10%, respectively.

Table 3. Results of instrumental variables.

Spatial weight matrix	Distance	Institution	Culture
Explained variable: urban overall innovation ability			
L centrality	−0.0578 (−0.69)	−0.1360** (−2.26)	−0.0437* (−1.76)
Explained variable: cooperative innovation ability			
L centrality	−0.0996 (−1.02)	−0.1432* (−1.70)	−0.0763** (−2.28)
Explained variable: independent innovation ability			
L centrality	−0.0393 (−0.52)	−0.1299** (−2.44)	−0.0361 (−0.41)

Standard errors are in parentheses. ***, ** and * indicates significant at the levels of 1%, 5%, and 10%, respectively.

greater advantage in attracting innovative talents and resources, which in turn encroaches on the innovation development of surrounding cities, resulting in negative externalities.

5. Conclusion and discussion

The Chinese government is trying to build GBA into an international polycentric innovation hub. Analysis in this paper shows that the knowledge innovation capability and network density of the GBA have continuously improved, but the four-centre innovation network pattern in the government plan has not yet been formed. Guangzhou and Hong Kong have always been innovation centres in this region. Since 2012, Shenzhen has become another innovation centre. However, Macao is not strong enough to be an innovation centre in the network. Although Hong Kong has superior innovation strength, under the influence of institutional and cultural differences, its innovation cooperation with mainland cities is relatively limited except for its cooperation with Guangzhou and Shenzhen. Most of the intercity innovation cooperation takes place within Guangdong province, especially between Guangzhou and Shenzhen. Consistent with the conclusion of other studies, this article also proves that there exists a multi-centre network structure in knowledge innovation in GBA (Ma and Zeng 2020). The difference of conclusion lies in that Hong Kong does not have a central position if technological innovation is considered (Li and Zhou 2020).

Results of spatial econometric models show that the impact of cities' innovation centrality on surrounding areas is negative. Small and medium-sized cities do not obtain borrowing size but are in the agglomeration shadow of central cities. In terms of influencing factors, the cultural and institutional differences between cities have far greater negative impacts on innovation than distance. Inspired by this, improving soft environment is a prerequisite for reducing the spatial gaps and promoting regional innovation. On the basis of the above conclusions, we propose the following suggestions for optimising the spatial effect of knowledge innovation network in GBA.

First, it is necessary to reduce the institutional barriers in innovation cooperation between mainland cities and Hong Kong/Macao. Hong Kong and Macau are special administrative regions in China with different political systems compared with the mainland. The current cross-border cooperation mechanism is not sound, which hinders the development of innovative cooperation and the free flow of elements. In this regard, city managers need to formulate more convenient policies from the aspects of personnel mobility, innovation resource sharing and cooperation projects to promote the development of innovation cooperation. For small and medium-sized cities, it is necessary to increase investment in knowledge innovation and institutional reforms to narrow the gap with central cities.

Second, create a more tolerant social environment that can be compatible with diverse cultures and personalities. The interests of immigrant should be safeguarded and make them enjoy the same treatment as the local residents, which is a key measure to eliminate the negative effects of cultural differences. Besides, improving the education level of residents will help to enhance the social inclusiveness and acceptance of cultural differences so as to exert the positive influence of cultural diversity on stimulating innovation.

Third, reducing the obstacles caused by inadequate infrastructure, such as transportation and Internet facilities, is still a task that cannot be ignored. Although great progress has been achieved in the construction of transportation networks in recent years, distance still has a certain negative impact on the flow of factors. Cities can jointly promote the construction of transportation in border areas by carrying out transportation cooperation projects. Among them, the intercity traffic between mainland cities and Hong Kong/Macao is the most important. In addition to facilitating the movement of people, construction of Internet facility is a precondition for information exchange and innovation cooperation, which is especially important for small and medium-sized cities.

It is worth noting that compared with Hong Kong, Guangzhou and Shenzhen, Macau's knowledge innovation ability is relatively weak. The development of higher education began late in Macau is one of the reasons for this phenomenon. Macau established its first modern university, the University of Macau, in 1981, which is far later than Hong Kong and Guangzhou. Given this, Macao should strengthen scientific research investment and talent exchange with Hong Kong and Guangzhou to strengthen its basic research.

Finally, strengthening the protection of property rights is an important measure to safeguard the legitimate interests of researchers and stimulate innovation. Hong Kong's experience and advantages in intellectual property protection should be fully utilised to establish a regional intellectual property protection and transaction centre. At the same time, an intellectual property information exchange mechanism and sharing platform should be established in the GBA to promote the legal and effective circulation of intellectual property.

Future research can be advanced from the following aspects. First, the measurement method of network structure can be improved. For example, locally weighted regression is a cutting-edge method that can accurately identify the centre of the network by fitting the urban density surface. Therefore, it can measure and identify urban innovation network centres more precisely. Second, the spatial effect of innovation network structure needs to be deeply analysed. In addition to influencing innovation itself, the innovation network structure will also have an impact on other aspects of the economy and society, as well as the urban spatial layout, which provides a wide range of topics for future research. Finally, the situation and mechanism of knowledge innovation and technological innovation are significantly different. Therefore, analysing the network externalities from the perspective of technological innovation is one of the contents of future work.

Note

1. In order to maintain the appropriate length, data description and statistical characteristics are not shown in the article.

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